COMPLETED

Seagull

Surf Life Saving: AI Swimmer Detection

A camera with an in-built artificial intelligence detection and tracking software to accurately and automatically locate swimmers in the rip currents to assist in Surf Lifesaving preventative operations.



15

Design Interations

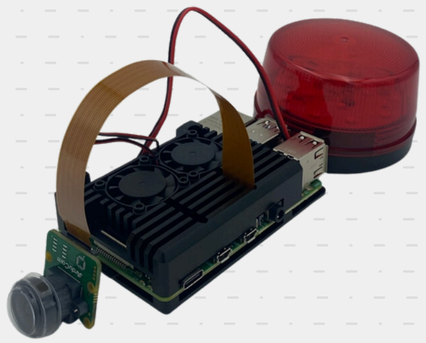
2nd

Place State ranking

1

Shape Exhibition Nomination

Design Iterations



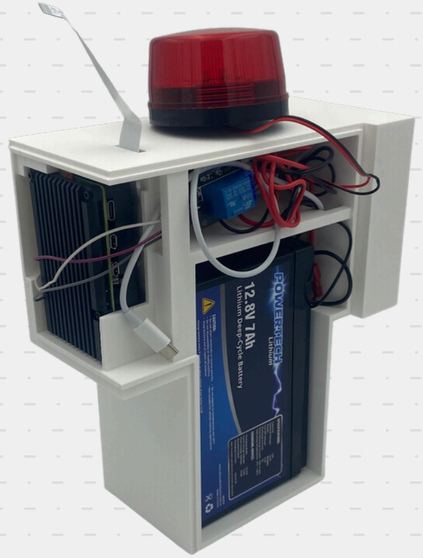
MARK 1

Main electrical components wired with uploaded AI swimmer detection software



MARK 2

Initial Rapsberry Pi and Camera dimesions gauged with laser cut plywood housing



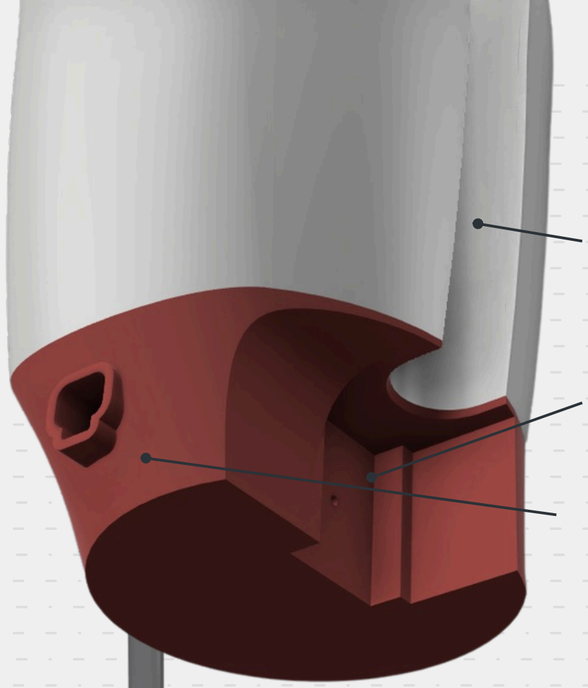
MARK 4

Housing module including battery, relay, transformer with counter balance on right for better balance on pole



MARK 15

Optimised housing module with waterproofing, shock resistant outershell and increased access for maintenance

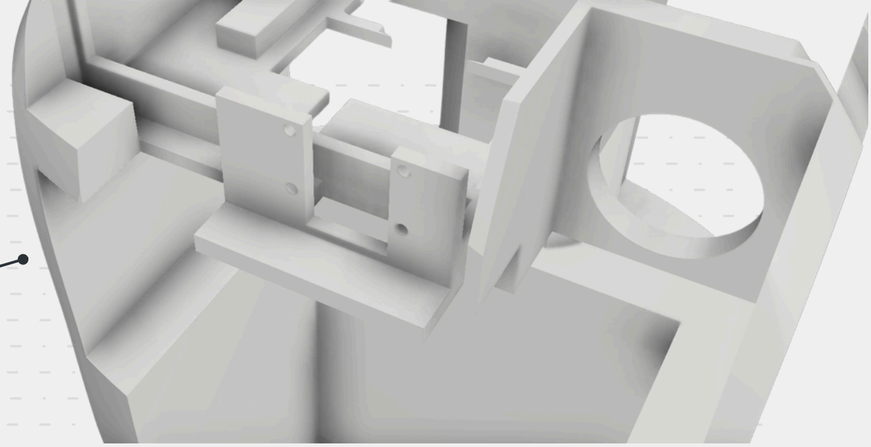


MARK 12

Slot for Surf Lifesaving poles for mounting stability

Area for GoPro universal mounting for easy attachment and operation

Water proof recharging port for battery



MARK 7

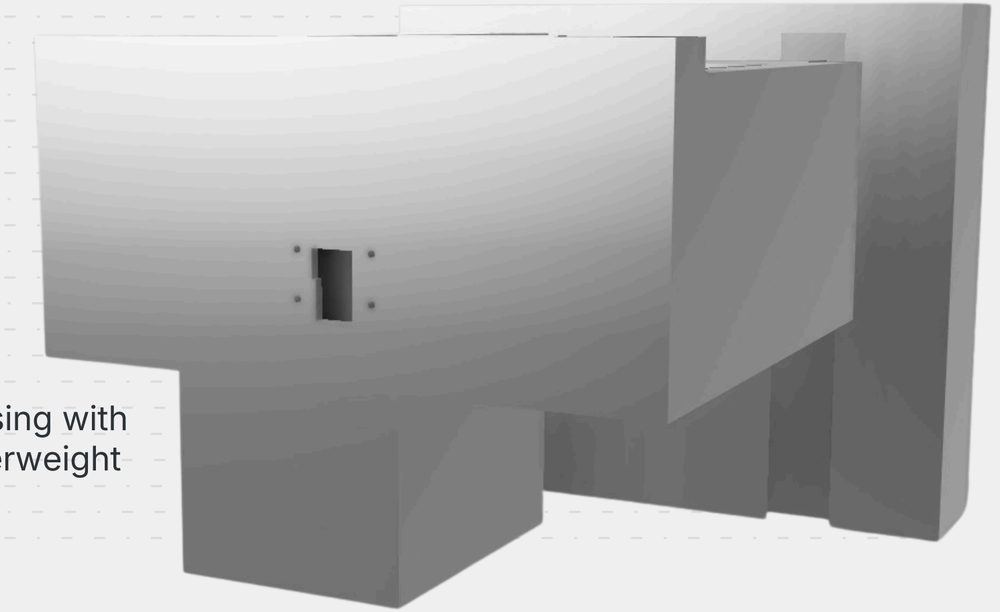
Interior Housing

Fusion 360 Models



MARK 15

Surf Lifesaving branding colours



MARK 3

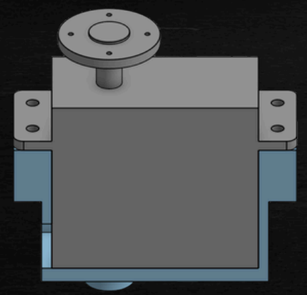
First 3D printed housing with dovetail back counterweight to attach to pole

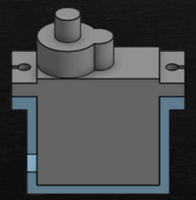
IN PROGRESS

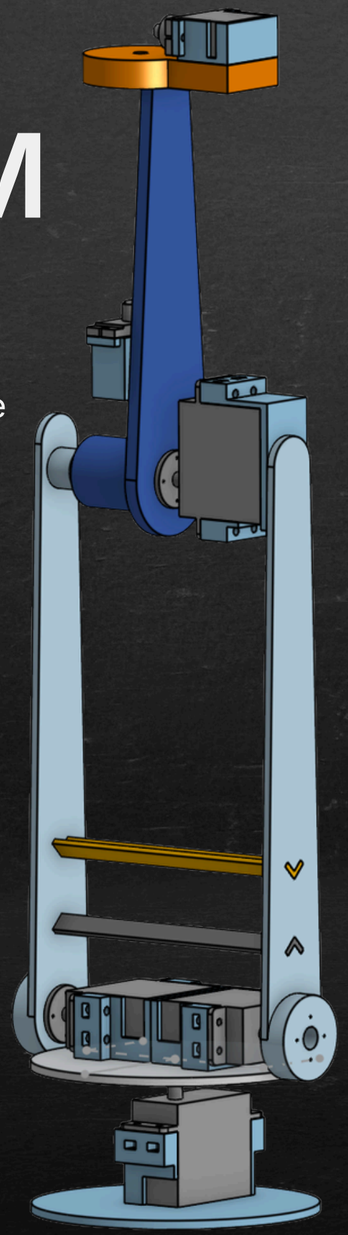
5 DOF ROBOTIC ARM

This robotic arm is to clear completed prints from my 3D printer and neatly place them on my desk. An alternative solution would be to design a simple “sweeper” however this would risk breaking delicate prints or muddle up print bed groups.

This is a self funded hobby project so all mechanical features will be aimed to be solved with 3D printed solutions or salvaged parts from household items and old projects.







Phase I (current phase) - physically building the arm with the MG955 and MG90S servo motors.

Phase II - writing the functioning code and calculating the inverse kinematics on Matlab.

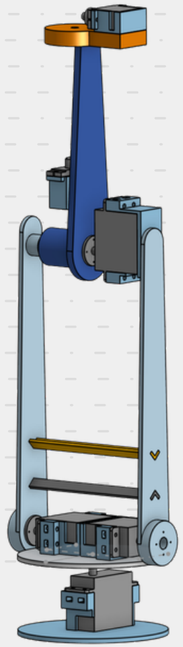
Phase III - adding a gripper as well as a camera and repurposed piezoelectric sensors in the gripper to allow for remote control.

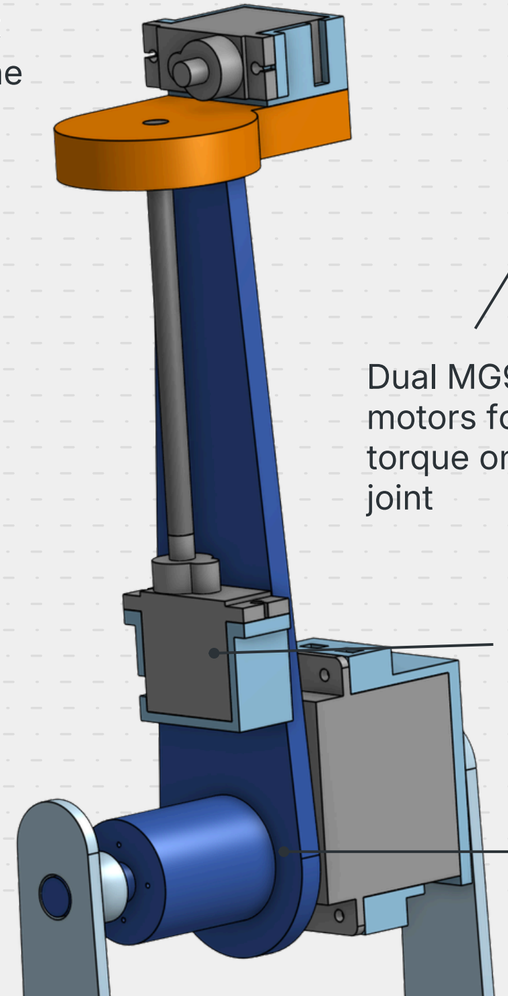
Phase IV - coding an app and UI so that the robot can be controlled remotely via a phone.

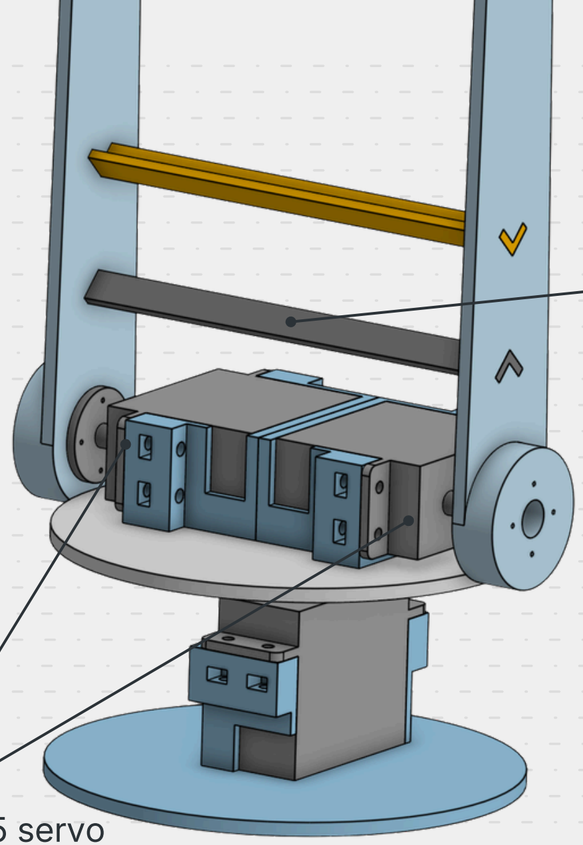
MARK I

Servo Orientation

This prototype aimed to work out the servo placement as well as the dimensions of all the arms and joints supported by moment calculations.







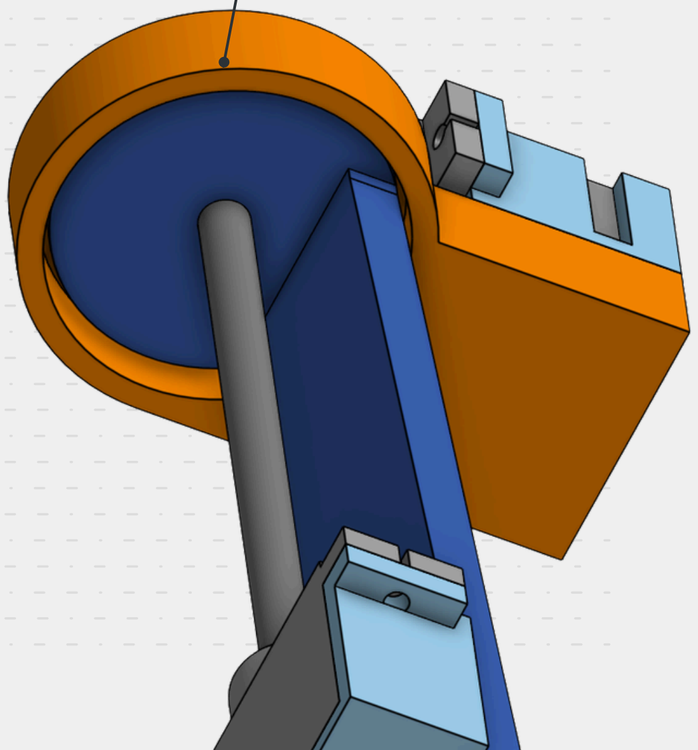
Arm braces to prevent unwanted movement of the arms but keep the parts to be 3D printed easily

Dual MG995 servo motors for increased torque on shoulder joint

Servo located lower on the forarm to reduce moment, whilst allowing for gripper rotation

Centrally balanced load

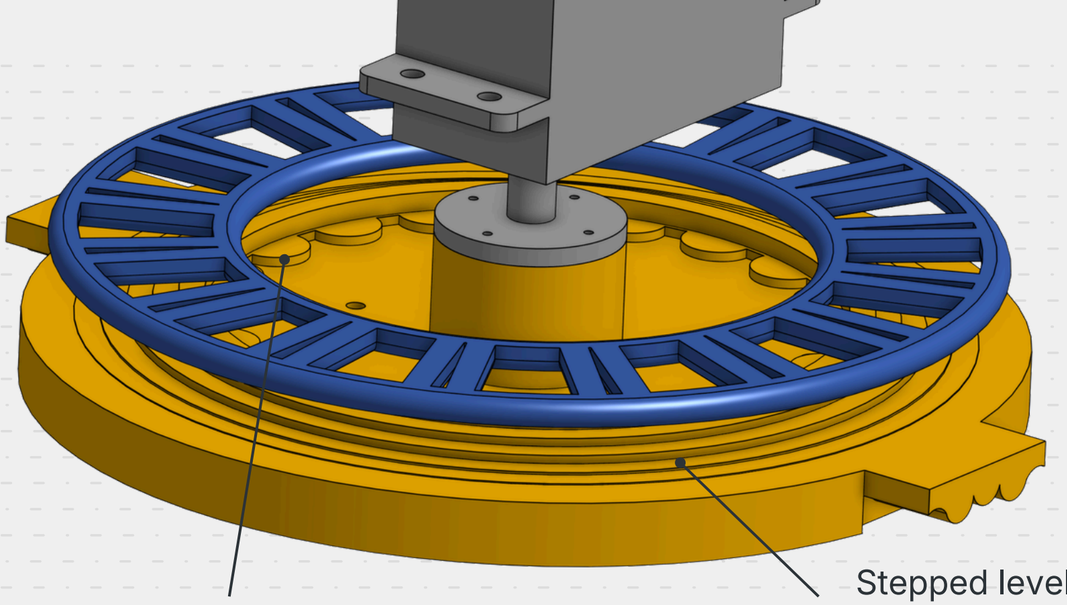
Gripper platform (**Orange**) rotates around forearm (**Blue**) powered by MG90S servo located on forearm



MARK 2

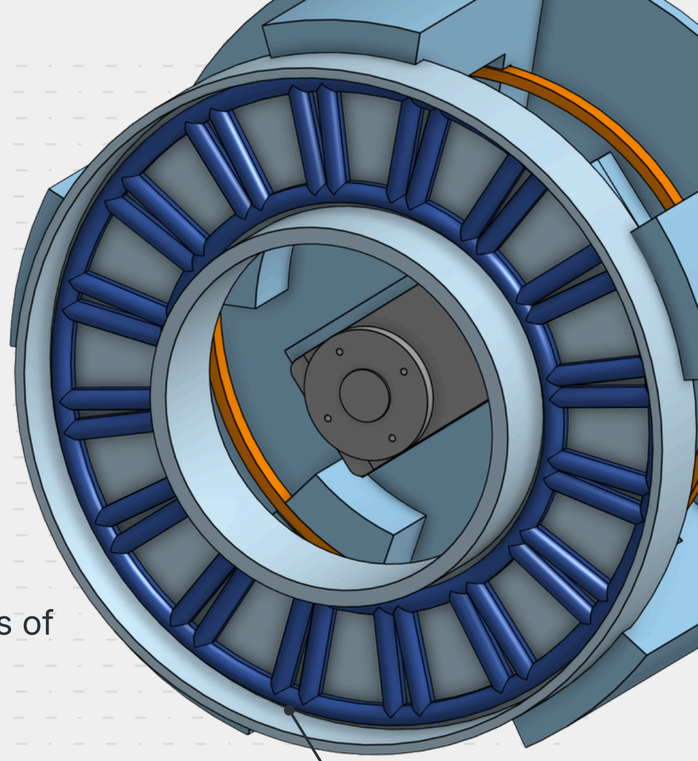
Part Support

This prototype aims to build viable shells and supports to house the servos mapped out in MARK I. Currently only the base and shoulder servo supports have been built.

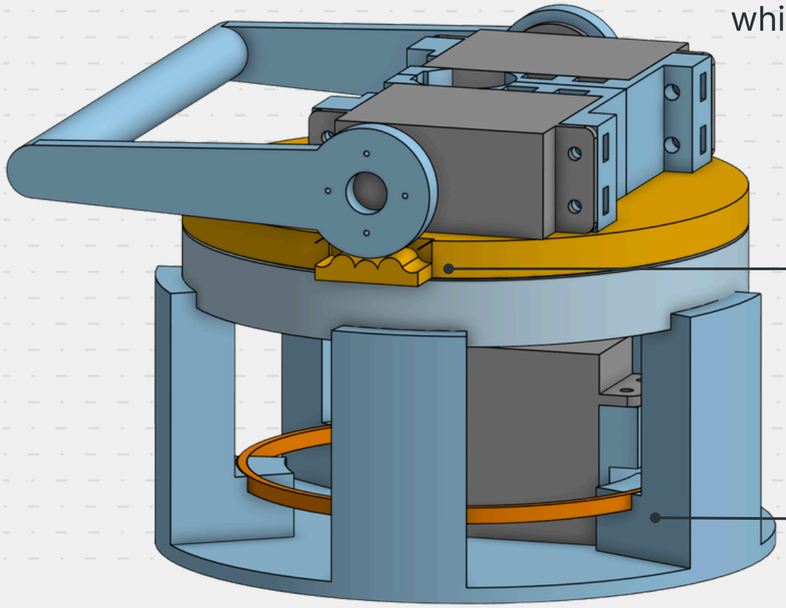


Circular offsets keeps the base (**Orange**) centred on the bearing housing (**Light Blue**) whilst minimising friction

Stepped levels allow base to have 5 points of contact with each truncated bearing



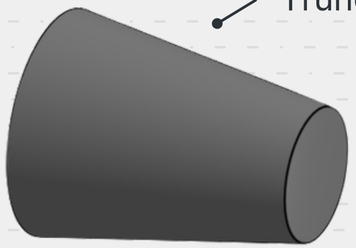
Bearing guide rail (**Blue**) keeps the truncated cone bearings in place

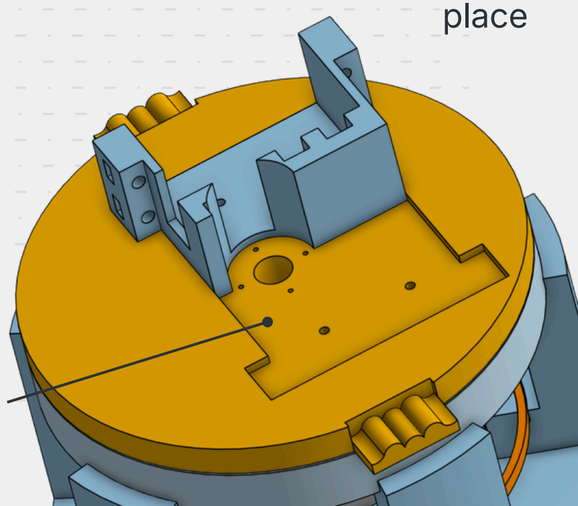


Upper arm support reduces shear forces on servo shaft. circular design minimises contact points and friction, whislt allowing smooth movement

Support ring to brace the support columns from internal forces

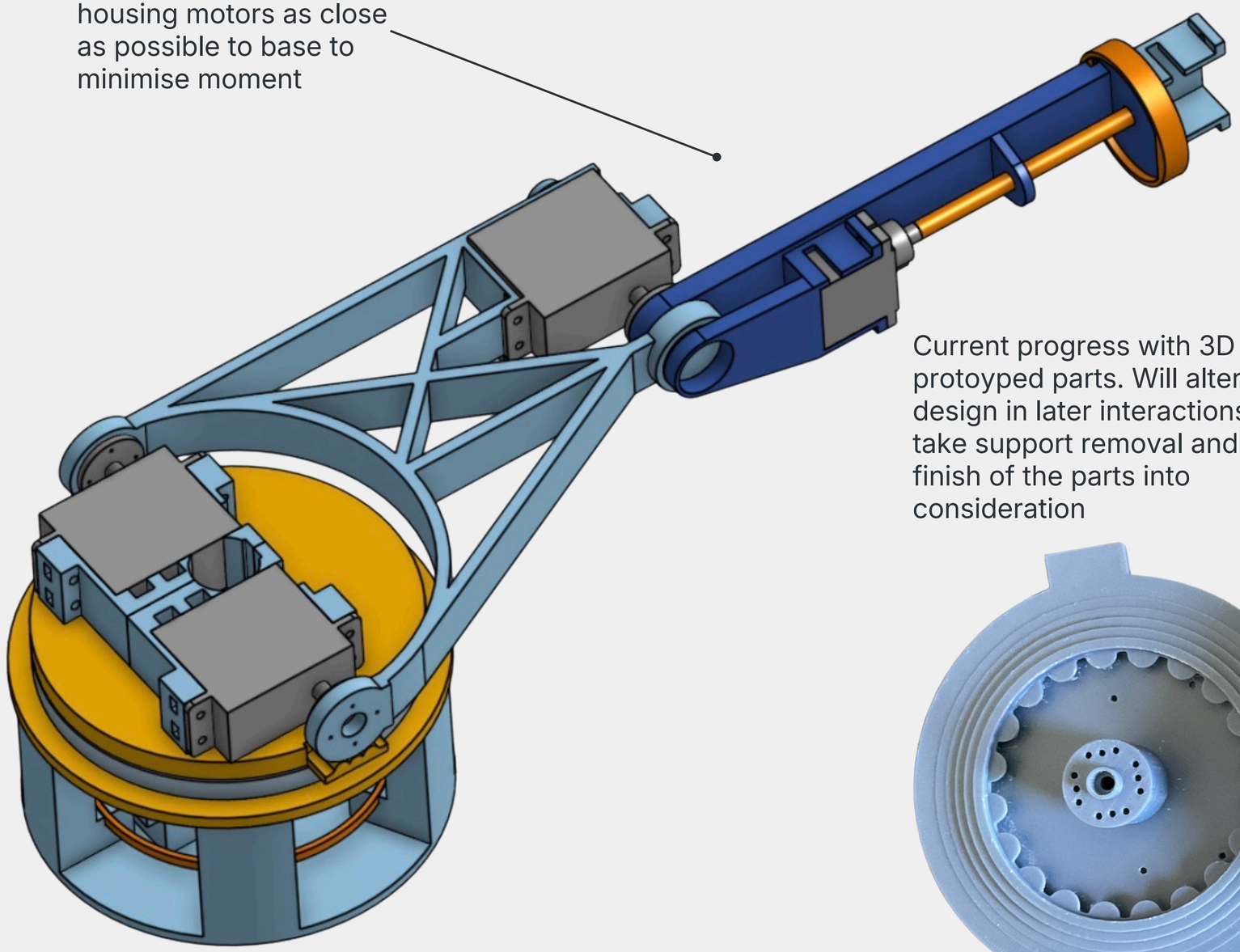
Truncated Bearing

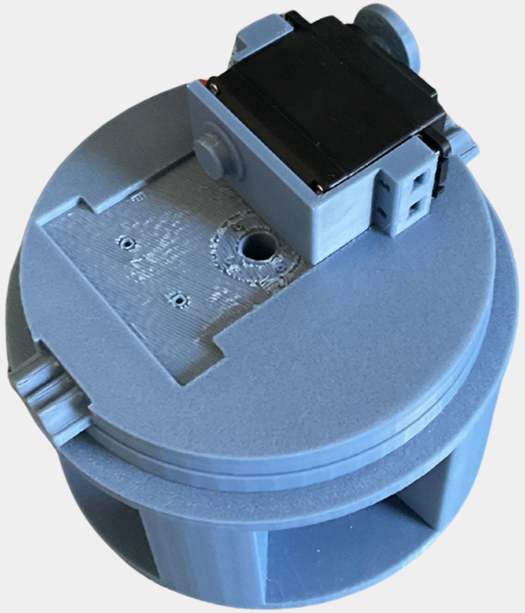




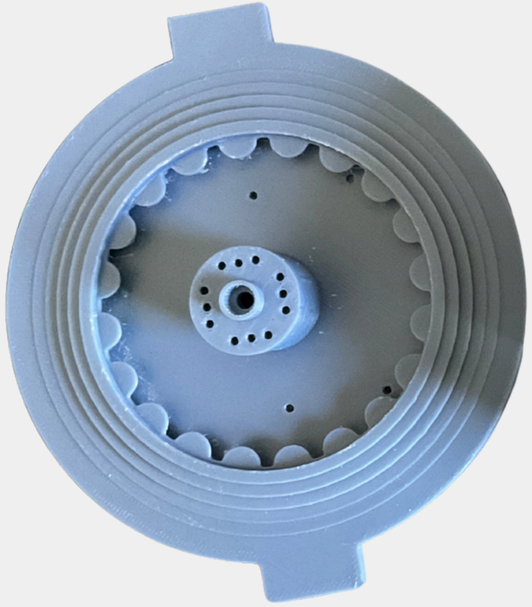
Servo housing indent on base allows for easy alignment whilst minimising 3D printing supports when printing the base

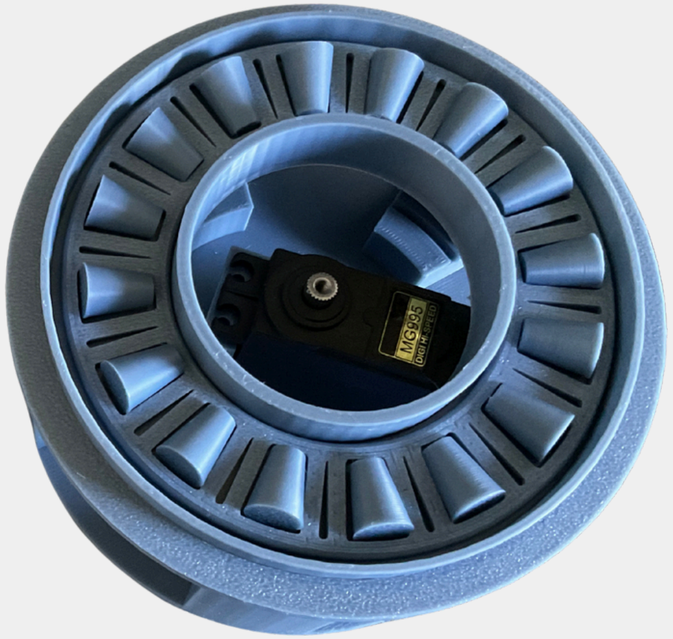
Current CAD design housing motors as close as possible to base to minimise moment





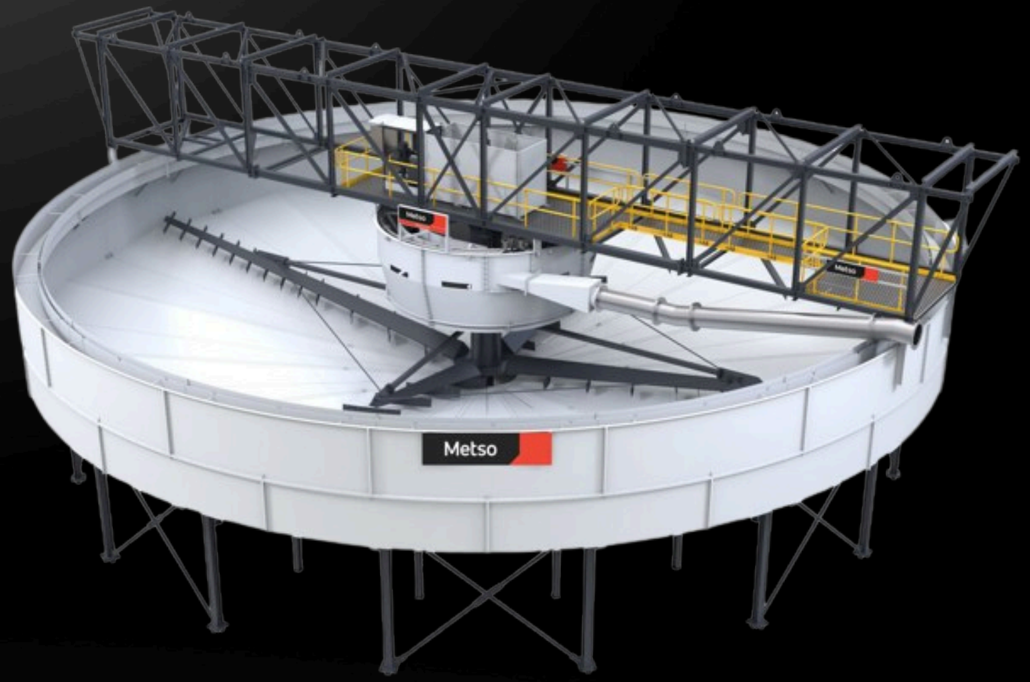
Current progress with 3D printed protoyped parts. Will alter the design in later interactions to take support removal and surface finish of the parts into consideration





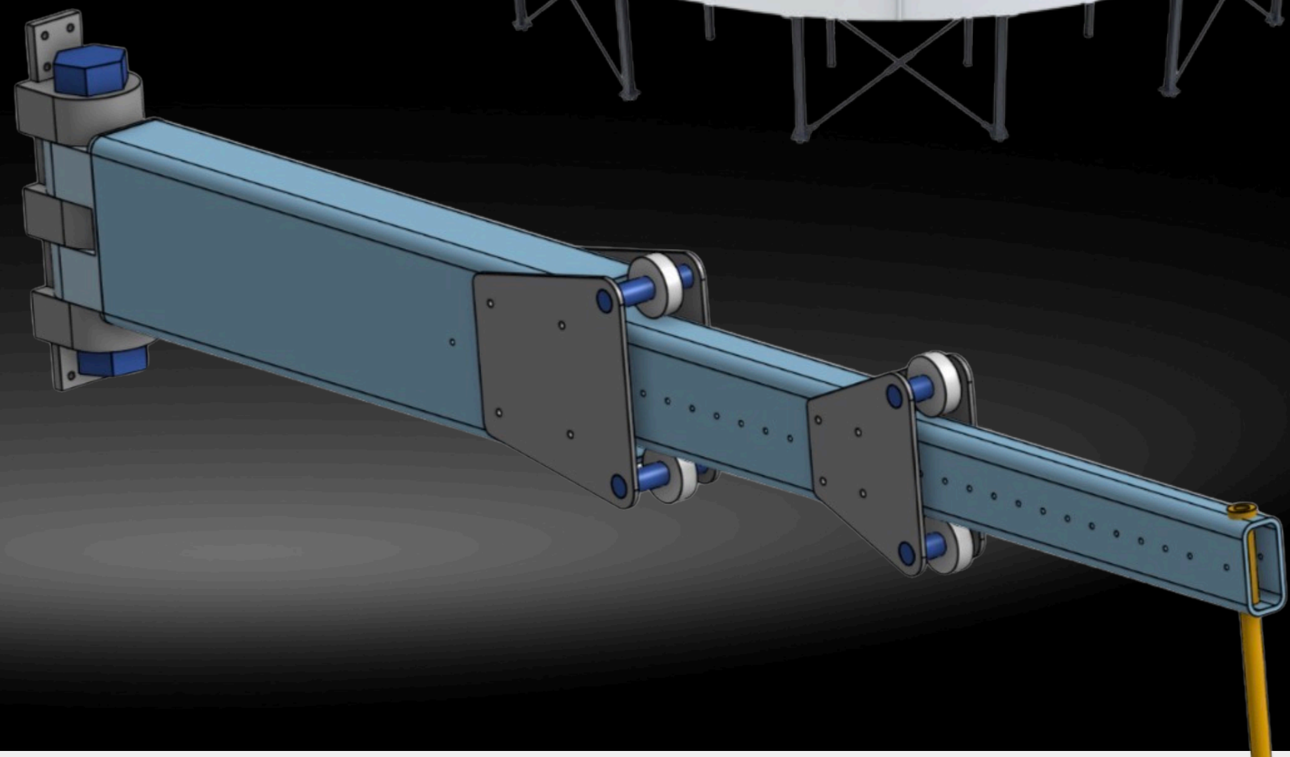
COMPLETED

Turbimax CUS50D Mounting Arm for Metso



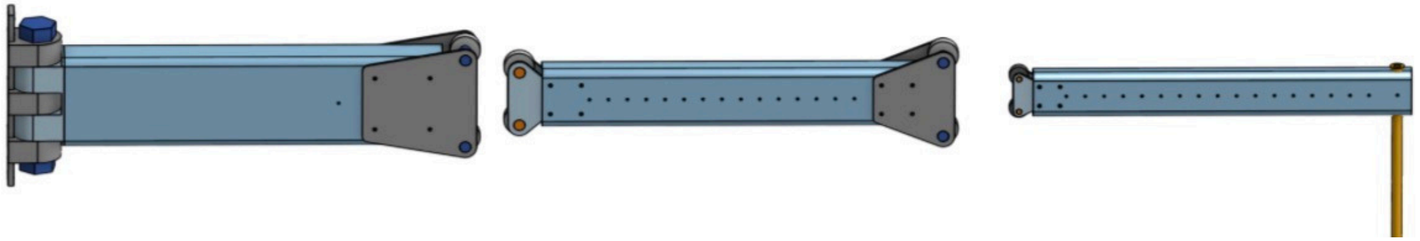
Designed a mounting system for an Endress+Hauser Turbimax turbidity sensor to be used on Metso thickeners. The mount was briefed to be attached to the central bridge along the thickener and be able to take turbidity measurements from any point along the edge.

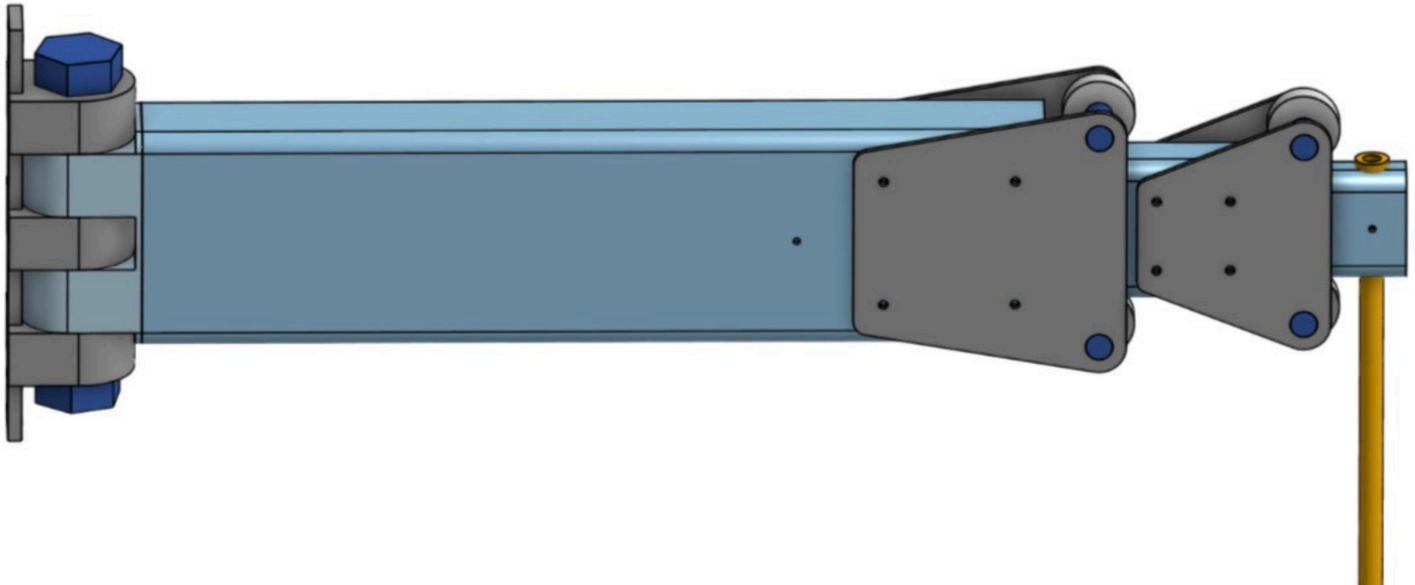
The final concept uses three telescoping RHS beams, a simple pivot joint, and dual spring-loaded detents. The key design focus was global manufacturability, by using standard hardware, and a fully parametrically driven CAD design so RHS and bearing specifications could be adapted to locally available components without redesigning the overall system.

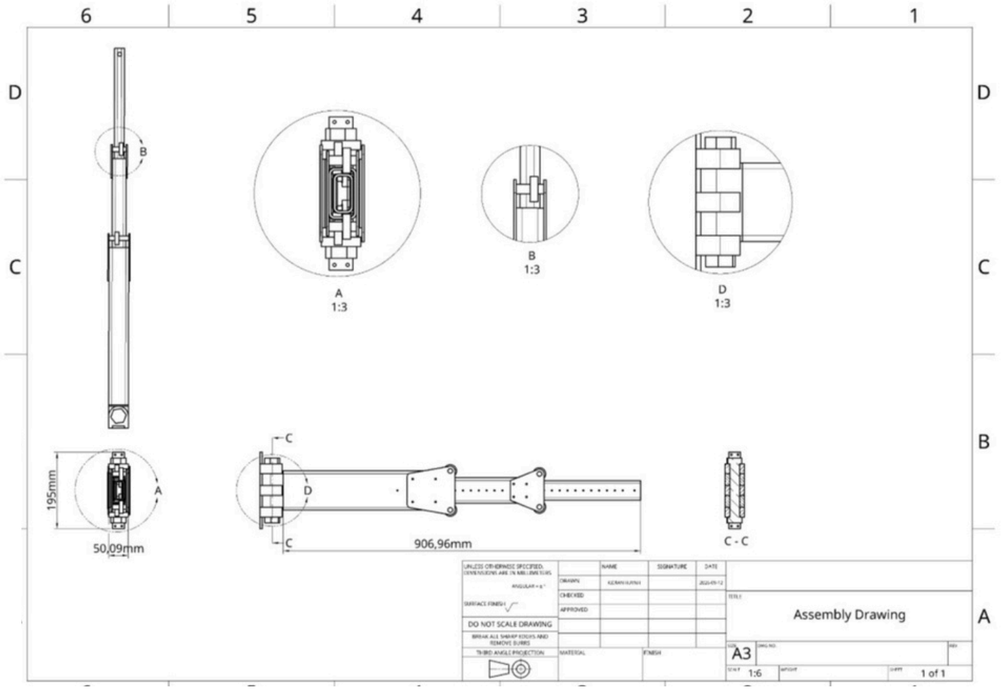


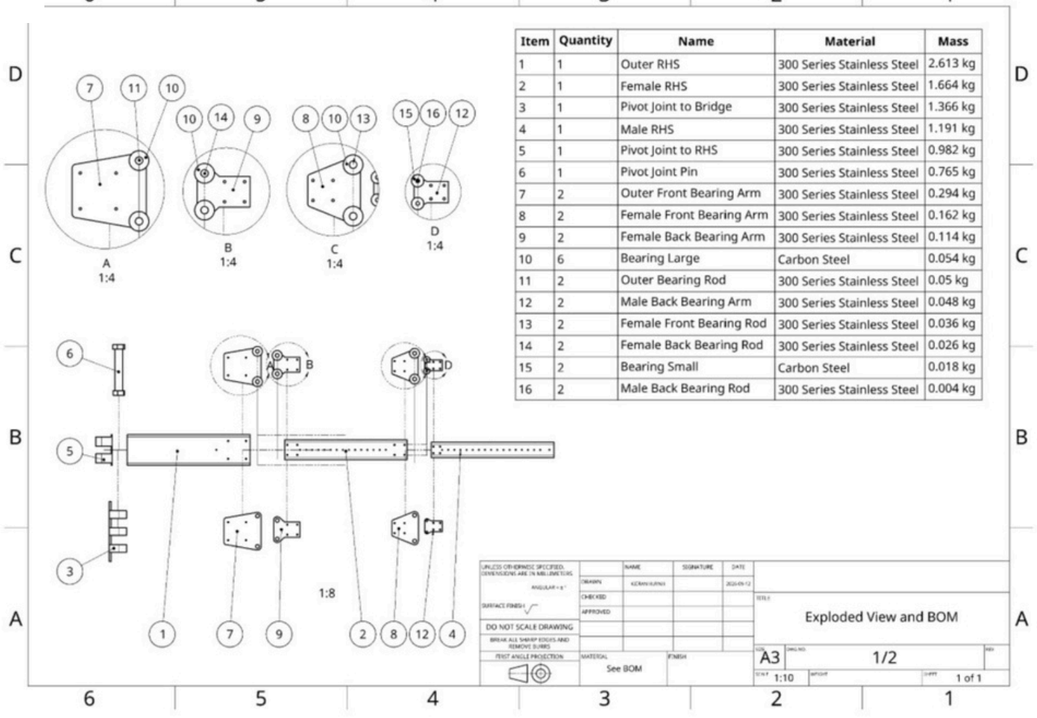
The project placed 3rd in the Metso designathon out of 60+ teams and 240+ students.

Engineering Drawings









Item	Quantity	Name	Material	Mass
1	1	Outer RHS	300 Series Stainless Steel	2.613 kg
2	1	Female RHS	300 Series Stainless Steel	1.664 kg
3	1	Pivot Joint to Bridge	300 Series Stainless Steel	1.366 kg
4	1	Male RHS	300 Series Stainless Steel	1.191 kg
5	1	Pivot Joint to RHS	300 Series Stainless Steel	0.982 kg
6	1	Pivot Joint Pin	300 Series Stainless Steel	0.765 kg
7	2	Outer Front Bearing Arm	300 Series Stainless Steel	0.294 kg
8	2	Female Front Bearing Arm	300 Series Stainless Steel	0.162 kg
9	2	Female Back Bearing Arm	300 Series Stainless Steel	0.114 kg
10	6	Bearing Large	Carbon Steel	0.054 kg
11	2	Outer Bearing Rod	300 Series Stainless Steel	0.05 kg
12	2	Male Back Bearing Arm	300 Series Stainless Steel	0.048 kg
13	2	Female Front Bearing Rod	300 Series Stainless Steel	0.036 kg
14	2	Female Back Bearing Rod	300 Series Stainless Steel	0.026 kg
15	2	Bearing Small	Carbon Steel	0.018 kg
16	2	Male Back Bearing Rod	300 Series Stainless Steel	0.004 kg

COMPLETED

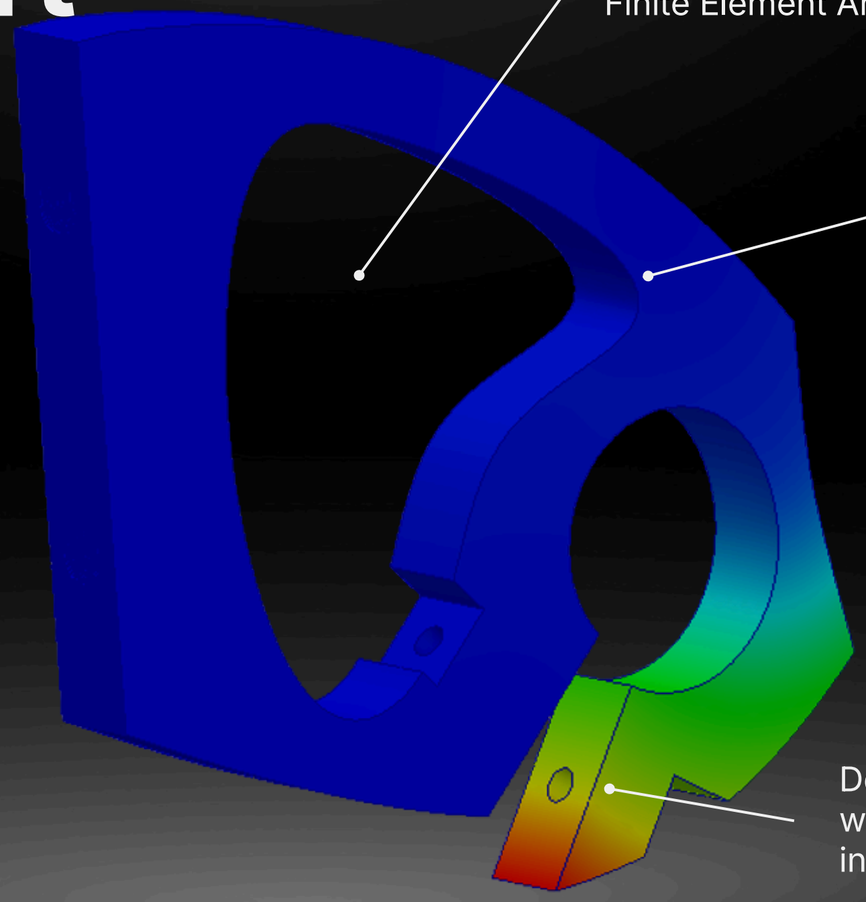
USYD Motorsport

FoS and Displacement simulations

This bracket was designed as a task for the USYD Motorsport Structures team to mount the jacking bar and withstand a 700N lifting load.

This bracket provides no advantages in performance however is regulated by FSAE competition rules. Thus, should minimal displacement and minimal weight.

This bracket was designed using an interactive process of FEA simulations then testing the changes with FoS and Displacement simulations.

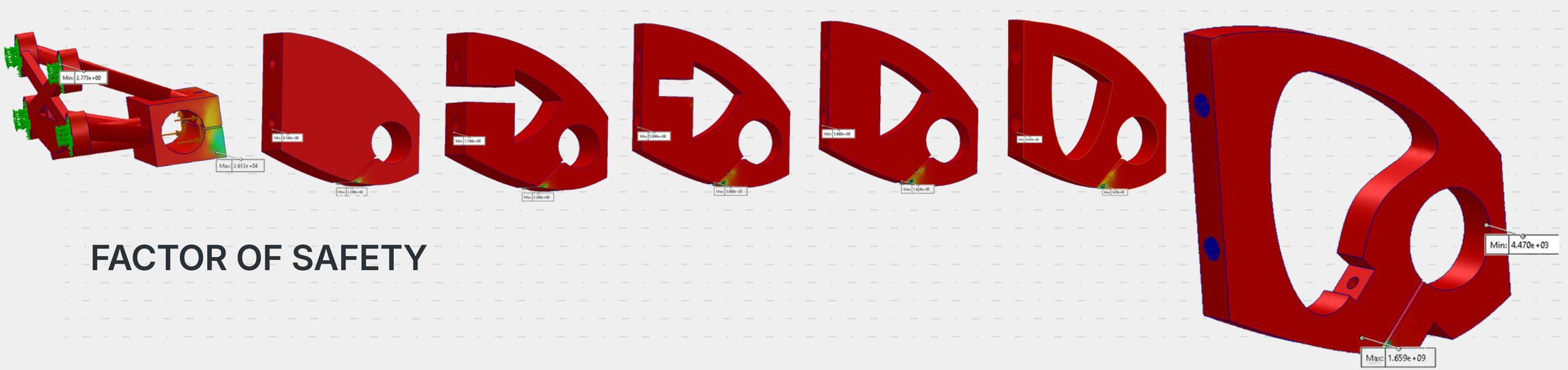


Solidworks Simulations

DISPLACEMENT



FACTOR OF SAFETY



COMPLETED

USYD Rocketry

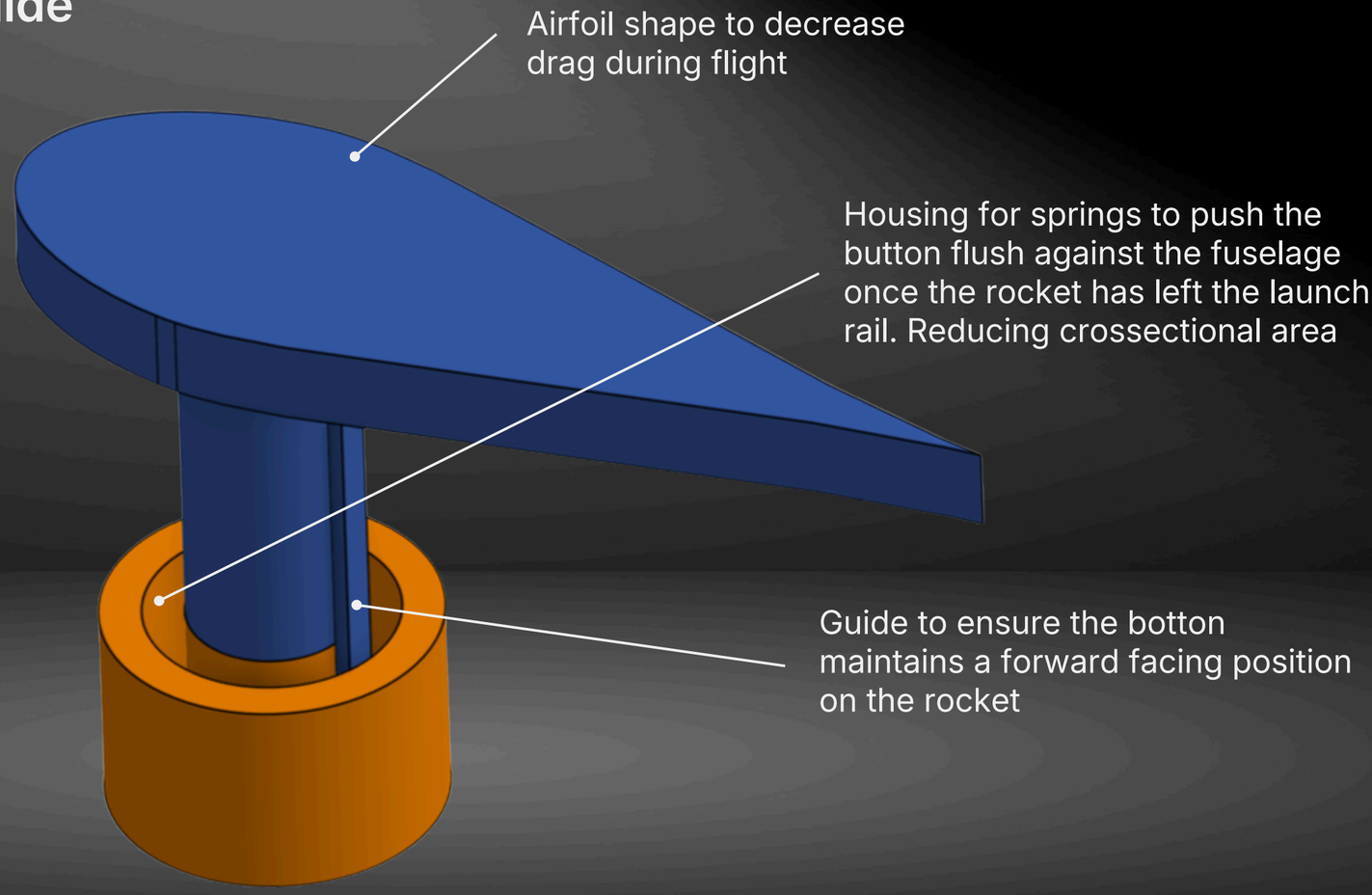
Aerodynamic Retractable Rail Guide

This rail guide button was designed as a task for the USYD Rocketry team.

A rocketry rail guide keeps a model or high-power rocket securely attached to a metal launch rail as it lifts off, ensuring it flies straight until it gains enough speed for its fins to work.

Issues with existing rocketry rail guides is that they cause considerable drag during flight limiting the rockets max apogee.

009



IN PROGRESS

BeerBound

IOS App for Tracking Nights out

BeerBound is an iOS app, coded on Swift, for tracking nights out with live GPS route recording, automatic stop detection, and photo to location matching using EXIF GPS. This app is currently in review by Apple before it is released to the App Store.

The app’s venue, deal, and event data is pulled with API flows from an external database created on Supabase, with offline caching for network failures.

The core engineering challenge was reliability under iOS's limits on running background processes. This included coded logic to ensure GPS fixes are filtered for accuracy, jitter, and long distance jumps caused by poor network connection on public transport and dense urban areas.

